

Biostat STR Microbial 50 L - 11-40 L working volume

sartorius-str-microbial-50

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	11 L	Source-supported
Maximum working volume	40 L	Source-supported
Nominal vessel / bag size	50 L	App assumption
Total vessel volume loaded into the form	50 L	Derived / estimated
Initial working volume loaded into the form	11 L	Derived / estimated
Product-contact material	Plastic / single-use	App assumption
Height : diameter	3	App assumption
Internal / chamber diameter	0.257 m	Derived / estimated
Liquid height at maximum working volume	0.771 m	Derived / estimated
Impeller / mixing type	Six-blade Rushton turbine	App assumption
Impeller count	3	App assumption
Impeller diameter	0.0848 m	Derived / estimated
Power number	5	App assumption
Initial agitation	250 rpm	App assumption
Maximum agitation	600 rpm	App assumption
Initial aeration	0.5 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0.1 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 11 L

Source-supported | Sartorius STR Microbial manufacturer technical article; Biostat STR Microbial - A Viable Single-Use Alternative

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 11 L.

Limitations: Direct manufacturer technical article; local raw HTML response retained. PDF datasheet remains unavailable.

[Local original: Sartorius STR Microbial manufacturer technical article - raw HTML](#)

[Manufacturer publication](#)

Maximum working volume - 40 L

Source-supported | Sartorius STR Microbial manufacturer technical article; Biostat STR Microbial - A Viable Single-Use Alternative

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 40 L.

Limitations: Direct manufacturer technical article; local raw HTML response retained. PDF datasheet remains unavailable.

[Local original: Sartorius STR Microbial manufacturer technical article - raw HTML](#)

[Manufacturer publication](#)

Nominal vessel / bag size - 50 L

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 50 L.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Total vessel volume loaded into the form - 50 L

Derived / estimated | Form-population rule

Rule: total volume = max(nominal vessel volume, maximum working volume)

Selection / calculation: max(50 L, 40 L) = 50 L

Limitations: This prevents the form's total vessel volume from being smaller than the working-volume ceiling.

Initial working volume loaded into the form - 11 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(11 L, 40 L) = 11 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Height : diameter - 3

App assumption | Vessel-specific preset override

Rule: heightDiameterRatio = 3

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This ratio is subsequently used to estimate vessel diameter and liquid height.

Internal / chamber diameter - 0.257 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \sqrt[3]{(4 \times V)/(\pi \times H:D)}$

Selection / calculation: $D = \sqrt[3]{(4 \times 0.04 \text{ m}^3)/(\pi \times 3)} = 0.2570098 \text{ m}$; the form displays 0.257 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 0.771 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.2570098 \text{ m} \times 3 = 0.7710293 \text{ m}$; the form displays 0.771 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Six-blade Rushton turbine

App assumption | Vessel-specific preset override

Rule: impellerType = "rushton"

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption.

Impeller count - 3

App assumption | Vessel-specific preset override

Rule: impellerCount = 3

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.0848 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter = $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation: $\max(0.33 \times 0.2570098 \text{ m}, 0.001 \text{ m}) = 0.08481322 \text{ m}$; the form displays 0.0848 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

Power number - 5

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].powerNumber` = 5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. The power number is an empirical generic value used in $P = N_p N_3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 250 rpm

App assumption | Vessel-specific preset override

Rule: $\text{baseRpm} = 250$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the initial agitation setting loaded into the form.

Maximum agitation - 600 rpm

App assumption | Vessel-specific preset override

Rule: $\text{maxRpm} = 600$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.5 vvm

App assumption | Vessel-specific preset override

Rule: $\text{baseVvm} = 0.5$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Vessel-specific preset override

Rule: $\text{maxVvm} = 2$

Selection / calculation: No calculation. The value is written directly into this vessel's catalogue entry.

Limitations: This vessel-specific value overrides the generic microbial stirred-tank profile. It is still treated as an app assumption unless the parameter is explicitly listed as source-supported. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].baffled = true`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].maxPressure = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.1 bar(g)

App assumption | Generic microbial stirred-tank profile

Rule: `PROFILE_DEFAULTS["microbial"].operatingPressure = 0.1`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable microbial stirred-tank starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

The microbial profile and Rushton-style mixing are source-aligned. Several control values remain editable simulator defaults.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Sartorius STR Microbial manufacturer technical article

sartorius-str-microbial-manufacturer-technical-article-1a11e44f9154.html.txt

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